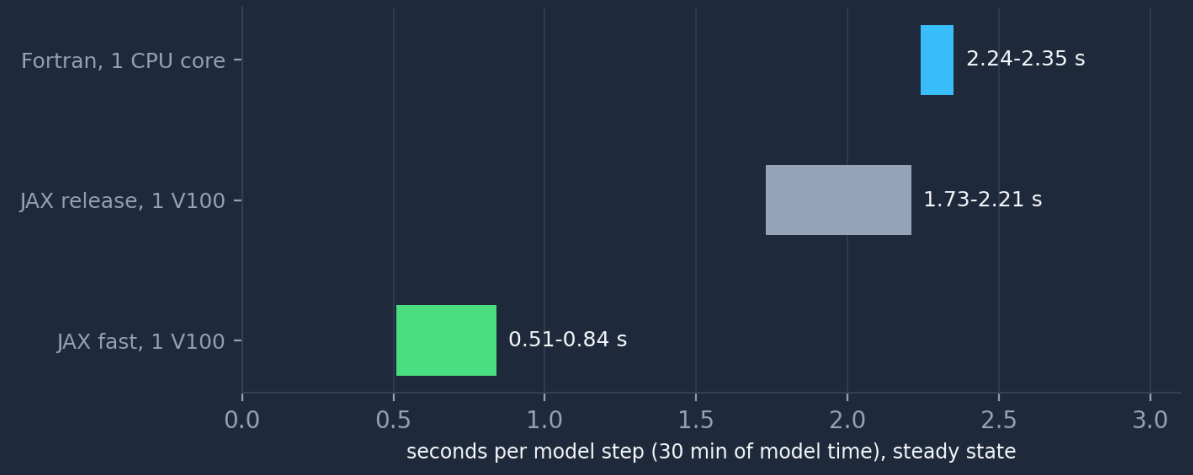
MANAGEMENT BRIEFING | 2026-10-19

# ROCKE-3D on JAX: where the project stands

Goal: run the ROCKE-3D P2SAoM40 climate model on GPUs through a JAX rewrite that is shown to behave like the original Fortran model.




Range over the runs on exclusive V100 nodes (Fortran on one CPU core). Steady state only: start-up (about 16 + 2 minutes of compilation) is excluded.

## 2.7-4.4x

**Fast configuration, steady state**

0.51-0.84 s per step against about 2.3 s for Fortran; the spread is unexplained.

## 0.75-0.97

**Release configuration / Fortran**

Near parity, not a speed win (1.7-2.2 s per step).

## NOT MET

**Strict fidelity rule**

Cloud ice at step 3 is 2.37x (release), 2.47x (fast) the reference spread; limit 2x.

**Bottom line: the JAX model runs a full day with radiation in JAX and is faster than Fortran in steady state, but it does not yet meet the agreed accuracy rule, the fast speed is not stable between jobs, and only the author has run the handoff. A research result, not a production claim.**

### Three labels to keep apart

RELEASE = all speed flags off, near parity. FAST = nine speed flags on, 2.7-4.4x steady state, unqualified. QUALIFICATION = repeat, fidelity, long-run and second-person checks still open.

Sources: STATUS_20261016 s5 (runs c-e and the 2026-10-18 step-time runs); RELEASE_NATIVE_RADIATION_20261012; ACCEPTANCE_CRITERIA s9-10

# How the project got here

Seven weeks, five stages: build a trustworthy reference, port piece by piece, couple, replace the radiation callback, then speed it up and hand it over.

### 1 REDUCED PORT

**28 Aug - 23 Sep**

Reduced driver with GPU kernels. It recommended against a full port; the owner reversed that.

**Kernels 2.3-6.3x faster than CPU (reduced scope, since superseded).**

### 2 REFERENCE + PIECES

**24 Sep - 6 Oct**

Real Fortran rebuilt as the oracle. Ocean, ice, dynamics, clouds and land ported and checked against it call by call.

**Every ported routine matched the real model to float64 precision; 680 tests passing by 30 Sep.**

### 3 COUPLE + SCORE A DAY

**6 - 8 Oct**

Pieces chained into one JAX-driven step; one model day scored against five real one-ulp runs.

**First coupled step 559 of 559 fields bitwise equal to the NumPy chain on 3 dates; day rule NOT MET.**

### 4 RADIATION + V100

**10 - 13 Oct**

SOCRATES radiation ported to JAX; first V100 benchmark; fast configuration built.

**Release near parity (0.95-0.97 of Fortran); fast 0.524 s/step in the archived run.**

### 5 HANDOFF

**16 - 19 Oct**

Minimal branch, one-command runner, shared environment, group-readable data, three briefings.

**Full runner: 7 jobs exit 0; 3,288 tests pass; only the author has run it.**

**Owner decisions that shaped it: end-to-end JAX first (7 Oct); a Fortran radiation callback was acceptable but must be labelled (7 Oct); the day must never exceed 2x the reference spread (8 Oct); a relaxed 3x rule may be used only for experiments and shown beside the strict result (11 Oct); amended criteria signed off (17 Oct).**

Sources: PROJECT_CHRONOLOGY (archive); Project_Summary_and_Conclusions; GOAL; ACCEPTANCE_CRITERIA s9-10, sign-off 2026-10-17; RELEASE_NATIVE_RADIATION_20261012

# What the evidence says, and does not say

The speed number is not the only gate: fidelity, repeatability and run length decide whether it is useful.

## SUPPORTED BY MEASUREMENTS

+ The full runner works end to end on this branch: seven SLURM jobs, all exit 0, summary table written (18 Oct).
+ Regression: 3,288 tests passed, 0 failed, 40 skipped, in both the release and the fast configuration.
+ Fast configuration: 0.51-0.53 s per step in four exclusive-node runs, which reproduces the archived 0.524 s.
+ Release configuration: near parity with Fortran on two nodes; native JAX radiation cut the median radiation step from 10.0 s to 1.9 s.
+ Day 2 of the scored window stays within 2x the reference spread (worst 1.22 release, 1.11 fast).

## NOT ESTABLISHED

+ Strict fidelity rule NOT MET on day 1: cloud ice at step 3 is 2.37x / 2.47x the spread (limit 2x), and steps 0-2 are further beyond. The relaxed experiment rule (3x from step 3) is met.
+ Fast speed is unstable: 0.51-0.53 s in four runs, 0.80-0.84 s in four others; the node, code, environment and Fortran runs were ruled out.
+ Runs are not bitwise repeatable (native radiation, same code).
+ One start date, one scored day, 108 timed steps; the 30-day run is invalid after step 1288.
+ Only the author has run the handoff, the shared environment and the shared data.

**Scope: nov26 start, V100, float64, inputs from one recorded Fortran run. Do not generalize to other seasons, GPUs or long production runs.**

Sources: STATUS_20261016 s2, s4, s5; ACCEPTANCE_CRITERIA s9-10; RELEASE_NATIVE_RADIATION_20261012; SPEED_STEP_TIME_LOG_20261012 (archive)

# Decisions needed from management

The next work should answer what the speedup is for and what 'good enough' means.

## 1 TARGET USE

+ What run length and restart cadence matter? Start-up costs about 17 minutes of compilation; payback against Fortran is about 12-15 model days (arithmetic, not measured).
+ Is the V100 the deployment target, or should another GPU be qualified?

## 2 SCIENCE BAR

+ Keep the strict 2x rule for release, or approve a separate experimental criterion?
+ Who owns the disposition of the cloud-ice excess at step 3 (2.37 -> 2.47 with the fast flags), and which further dates and run lengths are needed?

## 3 QUALIFICATION RESOURCES

+ Exclusive V100 time for repeats and for the step-time question.
+ A second developer to reproduce the handoff (nobody but the author has).

## 4 RELEASE CONTROL

+ The minimal branch is now the repository default; the full history is kept under a tag.
+ Turn the fast flags on by default only after the gates on the next slide pass. Support the shared Python environment and the Numba worker it needs.

**Recommended posture: fund a time-boxed qualification, not a production claim. Keep the release as the stable baseline.**

Sources: ACCEPTANCE_CRITERIA s9-10; STATUS_20261016 s4-6; SPEED_NEXT_STEPS_20261013 (archive); DATA_AND_PATHS

# Proposed gates: from a promising result to a decision

Small, ordered checks; stop or re-scope if the evidence does not support deployment.

| | | |
|---|---|---|
| A | **REPRODUCE** | A second person runs gpu/run_variants.sh from a fresh clone; explain why the fast step time is 0.52 s in some jobs and 0.80-0.84 s in others. |
| B | **QUALIFY THE BUILD** | Rebuild the environment from its file (untested); package the Numba worker; trace the 17 untraced test skips. |
| C | **CHECK FIDELITY** | Run the CPU day with Intel libimf; localize the repeat differences of native radiation; score more dates; keep strict and relaxed results side by side. |
| D | **LONGER RUNS** | Fix the 30-day failure (Fortran branch the port does not execute) and run the fast flags beyond two days. |
| E | **PROMOTE OR HOLD** | Compare the evidence with the target workload and the science bar, then decide flag defaults and release. |

**No schedule is implied: the owner sets dates, run counts and acceptance limits once the target workload is chosen.**

Sources: STATUS_20261016 s4-6; REPEATABILITY_AND_FLAGS_REGRESSION_20261014; QA_HANDOFF_20261016 (archive)